# Plastics, composites & elastomers

D&M WITH MR KELLY

Higher · Materials & manufacture · Worksheet

Name _____ Class _____ Date _____

**By the end of this worksheet I can…**

- ☐ explain the difference between thermoplastics and thermosetting plastics
- ☐ describe the properties and uses of ABS, acrylic, polypropylene and polystyrene
- ☐ describe the properties and uses of urea formaldehyde and melamine formaldehyde
- ☐ describe nylon, polythene and PVC
- ☐ describe elastomers and composites (GRP, carbon fibre)
- ☐ describe methods to identify materials, including what each result tells you
- ☐ explain how materials influence design and manufacture
- ☐ describe developments in new materials and their impact

## I DO — Worked example

Read how a full-mark answer is built

*A child's bike helmet.*

**1 Identify a suitable thermoplastic for the helmet shell and explain why it is suitable.** (3 marks)

**Full-mark answer** each ✓ is one mark

ABS ✓. It has high impact resistance, so it absorbs a knock without cracking and protects the child's head ✓. It is lightweight, so the helmet is comfortable to wear for long periods ✓.

**Examiner tip:** 2025 commentaries credited ABS or polypropylene with impact resistance, light weight, colours or scratch resistance.

**Why marks are not awarded**

*"Plastic, because it is strong."*

0 marks: no named thermoplastic and "strong" is not explained.

## WE DO — Guided practice

Use the starters and word bank to help you

*Swimming goggles have a silicone strap and seal.*

**2 Explain why an elastomer is suitable for the strap and seal.** (2 marks)

**Sentence starters**

One reason is … because …

Another reason is … so …

**Word bank**

against · comfortable · stretches · adjusts · flexible · shapes · tightly · different

Tip: 2 marks = 2 separate, developed points.

**YOU DO** **On your own** No notes. Check your answers afterwards.

### 3 Quick check: tick one box for each.

**a)** Which can be reheated and reshaped?
☐ Melamine formaldehyde ☐ Urea formaldehyde ☐ Acrylic ☐ Bakelite

**b)** Why is nylon used for small gears?
☐ It is transparent ☐ It is self-lubricating and wear resistant ☐ It is a thermoset ☐ It is cheap foam

**c)** A flip-top shampoo lid with a living hinge is made from...
☐ Acrylic ☐ Polypropylene ☐ Melamine ☐ Polystyrene

**d)** Which plastic is used for electrical plugs?
☐ Polystyrene ☐ Urea formaldehyde ☐ Acrylic ☐ Polypropylene

### 4 Key terms: write the letter of the matching description next to each term.

| | | |
|---|---|---|
| 1. **Thermoplastic** | ____ | **A** Sets permanently once heated; heat resistant; cannot be reshaped. |
| 2. **Thermosetting plastic** | ____ | **B** Softens when heated; can be reshaped and recycled. |
| 3. **ABS** | ____ | **C** Rigid, clear, polishes well. Glass substitute, signs. |
| 4. **Acrylic** | ____ | **D** Thermoset electrical insulator. Plugs and sockets. |
| 5. **Polypropylene** | ____ | **E** Tough, impact resistant, high gloss. Toys, casings, helmets. |
| 6. **Urea formaldehyde** | ____ | **F** Fatigue resistant (living hinges), chemical resistant. Caps, crates. |

### 5 Describe methods that could be used to identify the materials used in a product. (4 marks)

**6 Give the properties and one use of Melamine formaldehyde.** (2 marks)

**7 Give the properties and one use of Nylon.** (2 marks)

**Check and reflect**
Mark your answers with the answer sheet. Go back to the checklist on the site and re-rate each “I can” statement above. Which one still needs work?